

over the operating system's functionality. This facilitates adaptability for custom applications, industrial automation, and seamless functioning of defence software."

Contrary to popular belief, Maya OS isn't India's maiden attempt at an indigenously developed operating system. In 2007, C-DAC introduced the Bharat Operating System Solutions (BOSS) with the goal of promoting free and open source software adoption across India. BOSS, available in 19 Indian languages, was adopted by the Indian Army too.

The shift to Maya OS is primarily driven by the imperative to fend off cyberattacks and malware threats. Notably, the Indian Army faced a barrage of cyberattacks, with a report from 2019 indicating at least two attacks every month. Maya OS is poised to enhance security, cyber resilience, and diminish dependency on external systems. The heightened control over data and sensitive information is another boon that Maya OS promises to offer.

Biden Administration kicks off 'AI Cyber Challenge' to boost open source software safety

The White House has unveiled an ambitious initiative that aims to leverage the power of artificial intelligence (AI) in identifying and rectifying vulnerabilities within open source software. This groundbreaking effort, known as the 'AI Cyber Challenge', was announced by the Defense Advanced Research Projects Agency (DARPA) in August, signalling a new chapter in the battle against cyber threats.

The competition, set to be led by DARPA, carries significant incentives for participants. Teams entering the fray will vie for prizes totalling a staggering \$18.5 million. Additionally, the agency has allocated an extra \$7 million in prize



money specifically for small businesses that choose to partake in the challenge.

Central to the competition's objectives is the utilisation of AI technology to address software vulnerabilities, with a notable emphasis on open

source software. The Biden administration has garnered support from prominent AI industry players, including Anthropic, Google, Microsoft, and OpenAI, all of which have pledged to make their cutting-edge AI technology available for the challenge.

This strategic move by the White House and DARPA comes in response to escalating concerns surrounding cyber supply-chain risk, both within the federal government and the private sector. The initiative also aligns with the recent commitments made by seven leading AI companies in July, as they signed onto a framework of voluntary pledges brokered by the Biden administration.

The AI Cyber Challenge will proceed with a structured timeline. Participating teams will undergo a qualifying event in Spring 2024, with the top 20 scoring teams advancing to the semi-finals at DEFCON 2024. From these, up to five teams will receive monetary prizes and move on to the final phase of the competition, scheduled to be held at the annual DEFCON Hacker convention in 2025.

Nephio R1 launched to simplify cloud-native automation for telcos

The Nephio project, announced by the Linux Foundation over a year ago, is making headway with its maiden major release, Nephio R1. This collaborative endeavour aims to simplify cloud-native automation for telecommunication companies.

The project leverages Kubernetes as a foundational layer for the seamless management of cloud infrastructure and network functions, catering to the evolving demands of 5G operations.

India to back open source RISC-V architecture

India has thrown its support behind the open source RISC-V architecture. This move comes as RISC-V gains momentum as a potent contender on the global stage, challenging the dominance of Softbank's ARM technology.

Under the banner of the Digital India RISC V (DIR-V) programme, India's computing systems are set to harness the power of this open source architecture. The country's ambitions span various sectors including automotive, space technology, IoT sensors, and mobility, as highlighted by Rajeev Chandrasekhar, the minister of state for electronics and information technology.

Speaking about the initiative, Chandrasekhar affirmed, "The government is committed to making DIR-V the Indian ISA. There were some doubts if we will go the ARM way, but I want to commit to you today that DIR-V will have a central place in the high performance computing systems being developed by our country." The minister delivered this announcement virtually at the Digital India RISC-V Symposium, jointly organised by the Ministry of Electronics and Information Technology (MeitY) and IIT-Madras in Chennai.



The adoption of the open source RISC-V architecture not only signals India's aspirations for self-reliance but also secures its place at the forefront of global chip design developments.

As India takes its stand alongside the growing RISC-V community, it is clear that the nation is ready to make its mark in the semiconductor landscape.